

Homework 2

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Exercise 1

Since $Z_1, \dots, Z_R$ are independent, we obtain

$$\begin{aligned}\text{Var } \hat{z} &= \frac{1}{R^2} \sum_{r=1}^R \text{Var } Z_r = \frac{\sigma^2}{R}, \\ \text{E}\hat{z} &= \frac{1}{R} \sum_{r=1}^R \text{E}Z_r = z, \\ \text{E}\hat{z}^2 &= \text{Var}\hat{z} + (\text{E}\hat{z})^2 = \frac{\sigma^2}{R} + z^2, \\ \text{E}(\hat{z} - z)^2 &= \text{E}\hat{z}^2 - 2z\text{E}\hat{z} + z^2 = \frac{\sigma^2}{R},\end{aligned}\tag{1}$$

by substitution of the above results into each other. Consider the expected value of the unbiased sample mean,

$$\begin{aligned}\text{E}S^2 &= \text{E} \frac{1}{R-1} \sum_{r=1}^R (Z_r - \hat{z})^2 = \text{E} \frac{1}{R-1} \sum_{r=1}^R ((Z_r - z) - (\hat{z} - z))^2 \\ &= \text{E} \frac{1}{R-1} \sum_{r=1}^R (Z_r - z)^2 - 2(Z_r - z)(\hat{z} - z) + (\hat{z} - z)^2 \\ &= \text{E} \left(\frac{1}{R-1} \sum_{r=1}^R (Z_r - z)^2 - \frac{2}{R-1} (\hat{z} - z) \sum_{r=1}^R (Z_r - z) + \frac{R}{R-1} (\hat{z} - z)^2 \right).\end{aligned}$$

Notice that

$$\hat{z} - z = \frac{1}{R} \sum_{r=1}^R Z_r - z = \frac{1}{R} \sum_{r=1}^R (Z_r - z).$$

Hence,

$$R(\hat{z} - z) = \sum_{r=1}^R (Z_r - z).$$

Therefore, using Equation (1) and the definition of variance,

$$\begin{aligned}
\mathbb{E}S^2 &= \mathbb{E} \left(\frac{1}{R-1} \sum_{r=1}^R (Z_r - z)^2 - \frac{2R}{R-1} (\hat{z} - z)^2 + \frac{R}{R-1} (\hat{z} - z)^2 \right) \\
&= \mathbb{E} \left(\frac{1}{R-1} \sum_{r=1}^R (Z_r - z)^2 - \frac{R}{R-1} (\hat{z} - z)^2 \right) \\
&= \frac{1}{R-1} \sum_{r=1}^R \mathbb{E}(Z_r - z)^2 - \frac{R}{R-1} \mathbb{E}(\hat{z} - z)^2 \\
&= \frac{R\sigma^2}{R-1} - \frac{R}{R-1} \frac{\sigma^2}{R} = \sigma^2.
\end{aligned}$$

This means that the unbiased version of sample mean S^2 is indeed unbiased.

The bias of the biased version of sample mean is

$$\sigma^2 - \frac{R-1}{R} \mathbb{E}S^2 = \sigma^2 - \left(1 - \frac{1}{R}\right) \sigma^2 = \frac{\sigma^2}{R}.$$

Exercise 2

Errors We know that the estimate

$$\hat{z} \in \left(\pi - 1.96 \frac{\sqrt{2.70}}{\sqrt{R}}, \pi + 1.96 \frac{\sqrt{2.70}}{\sqrt{R}} \right).$$

with probability 0.95. If we want to get the leading figure 3 of π right, i.e. $3 \leq \hat{z} < 4$ w.p. 0.95, we must have that

$$\begin{aligned}
3 &\leq \pi - 1.96 \frac{\sqrt{2.70}}{\sqrt{R}}, \\
\pi + 1.96 \frac{\sqrt{2.70}}{\sqrt{R}} &\leq 4.
\end{aligned}$$

It follows that $R \approx 518$. Similarly, to get the 3.1 correct, $R \approx \frac{1.96^2 2.7}{0.0415} \approx 6023$. Finally, to get the 3.14 correct w.p. 0.95, $R \approx \frac{1.96^2 2.7}{0.0015} \approx 4609920$.

In Example 1.2, the first error is the squared variance 2.70, and the second error is the inaccuracy 0.5 because if the accuracy 0.5 is permitted, then it is possible that $\hat{z} = 3.14 - 0.5 = 2.64$, which does not have a leading 3 figure as required.

Exercise III.1.2 Notice that

$$\hat{p} = \frac{1}{R} \sum_{r=1}^R Z_r$$

where Z_r iid $\sim$ Bernoulli($\frac{2L}{\pi}$). Since $\sigma^2 = \text{var}(Z_r) = \frac{2L}{\pi} (1 - \frac{2L}{\pi}) < \infty$, by CLT, w.p. 0.95

$$\hat{p} \in \left(p - 1.96 \frac{\sigma}{\sqrt{R}}, p + 1.96 \frac{\sigma}{\sqrt{R}} \right).$$

We know that $\hat{p} = \frac{2L}{\pi}$. Hence, w.p. 0.95

$$\frac{2L}{\hat{\pi}} \in \left(\frac{2L}{\pi} - 1.96 \frac{\sigma}{\sqrt{R}}, \frac{2L}{\pi} + 1.96 \frac{\sigma}{\sqrt{R}} \right).$$

Multiplying by $\pi \hat{\pi}$ yields that

$$2L\pi \in \left(\hat{\pi}(2L - 1.96\pi \frac{\sigma}{\sqrt{R}}), \hat{\pi}(2L + 1.96\pi \frac{\sigma}{\sqrt{R}}) \right).$$

Assume

$$R > \frac{1.96^2(2L\pi - 4L^2)}{4L^2}. \quad (2)$$

Then $2L - 1.96\pi \frac{\sigma}{\sqrt{R}} > 0$, and w.p. 0.95

$$\hat{\pi} \in \left(\frac{2L\pi}{2L + 1.96\pi \frac{\sigma}{\sqrt{R}}}, \frac{2L\pi}{2L - 1.96\pi \frac{\sigma}{\sqrt{R}}} \right) = \left(\frac{2}{2 + 1.96\pi \frac{\sigma}{L\sqrt{R}}} \pi, \frac{2}{2 - 1.96\pi \frac{\sigma}{L\sqrt{R}}} \pi \right).$$

For greater efficiency, choose L that minimizes $\pm 1.96\pi \frac{\sigma}{L\sqrt{R}}$, which is equivalent to minimizing $\frac{1}{L} \sqrt{\frac{2L}{\pi} (1 - \frac{2L}{\pi})}$. It follows that $L = 1$, or as close to 1 as possible, is optimal. For the assumption (2), it means that for $L = 1$, $R > 3$.

We obtain that for $L = 1$

$$\hat{\pi} \in \left(\frac{2\pi}{2 + \frac{1.96}{\sqrt{R}} \sqrt{2\sqrt{\pi} - 2}}, \frac{2\pi}{2 - \frac{1.96}{\sqrt{R}} \sqrt{2\sqrt{\pi} - 2}} \right)$$

w.p. 0.95. If we want that $x \leq \hat{\pi} < y$ w.p. 0.95, then

$$x \leq \frac{2\pi}{2 + \frac{1.96}{\sqrt{R}} \sqrt{2\sqrt{\pi} - 2}}$$

and

$$\frac{2\pi}{2 - \frac{1.96}{\sqrt{R}} \sqrt{2\sqrt{\pi} - 2}} < y.$$

Using Assumption (2), we solve for R and obtain that

$$\sqrt{R} \geq \frac{1.96x \sqrt{2(\pi - 2)}}{2\pi - 2x}$$

and

$$\sqrt{R} > \frac{1.96y \sqrt{2(\pi - 2)}}{2y - 2\pi}.$$

For example, to get the leading 3 in π correct w.p. 0.95, we must have $3 \leq \hat{\pi} < 4$, which gives that $R \approx 985$. To get the leading 3.1 correct, $R \approx 12181$, and to get the leading 3.14 correct, $R \approx 8523340$.

Exercise 3

If X and Y are independent, then $\text{Var}XY = EX^2EY^2 - (EX)^2(EY)^2$. Since $\text{Var}X = EX^2 - (EX)^2$, $EX^2 = \text{Var}X + (EX)^2$. Therefore,

$$\begin{aligned}\text{Var}XY &= (\text{Var}X + (EX)^2)(\text{Var}Y + (EY)^2) - (EX)^2(EY)^2 \\ &= (EX)^2\text{Var}Y + (EY)^2\text{Var}X + \text{Var}X\text{Var}Y.\end{aligned}\quad (3)$$

Notice that

$$\text{Var} \frac{1}{R} \sum_{r=1}^R Z_r'' = \frac{1}{R} \sum_{r=1}^R \text{Var} Z_r'' = \frac{\sigma''^2}{R}.\quad (4)$$

By independence and Identity (3) and (4),

$$\begin{aligned}\text{Var} \left(\frac{1}{R} \sum_{r=1}^R Z_r' \frac{1}{R} \sum_{l=1}^R Z_l'' \right) &= \left(E \frac{1}{R} \sum_{r=1}^R Z_r' \right)^2 \text{Var} \frac{1}{R} \sum_{l=1}^R Z_l'' \\ &\quad + \left(E \frac{1}{R} \sum_{l=1}^R Z_l'' \right)^2 \text{Var} \frac{1}{R} \sum_{r=1}^R Z_r' \\ &\quad + \text{Var} \frac{1}{R} \sum_{r=1}^R Z_r' \text{Var} \frac{1}{R} \sum_{l=1}^R Z_l'' \\ &= (z')^2 \frac{\sigma''^2}{R} + (z'')^2 \frac{\sigma_r^2}{R} + \frac{\sigma_r^2 \sigma''^2}{R^2}.\end{aligned}$$

Similarly,

$$\begin{aligned}\text{Var} \frac{1}{R} \sum_{r=1}^R Z_r' Z_r'' &= \frac{1}{R^2} \sum_{r=1}^R \text{Var} Z_r' Z_r'' \\ &= \frac{1}{R^2} \sum_{r=1}^R (E Z_r')^2 \text{Var} Z_r'' + (E Z_r'')^2 \text{Var} Z_r' + \text{Var} Z_r' \text{Var} Z_r'' \\ &= (z')^2 \frac{\sigma''^2}{R} + (z'')^2 \frac{\sigma_r^2}{R} + \frac{\sigma_r^2 \sigma''^2}{R}.\end{aligned}$$

Hence,

$$\text{Var} \frac{1}{R} \sum_{r=1}^R Z_r' Z_r'' - \text{Var} \left(\frac{1}{R} \sum_{r=1}^R Z_r' \frac{1}{R} \sum_{l=1}^R Z_l'' \right) = \frac{R\sigma_r^2 \sigma''^2 - \sigma_r^2 \sigma''^2}{R^2} \geq 0,$$

which means that the first estimator has the smaller variance.

By Identity (4), we see that the mean $\frac{1}{R} \sum_{l=1}^R Z_l''$ has variance $\frac{\sigma''^2}{R}$, which is lower than the variance σ''^2 of Z_r'' . Notice the similarity

$$\frac{1}{R} \sum_{r=1}^R Z_r' \frac{1}{R} \sum_{l=1}^R Z_l'' = \frac{1}{R} \sum_{r=1}^R \left(\frac{1}{R} \sum_{l=1}^R Z_l'' \right) Z_r'$$

between the two estimators. Multiplication of each replicate Z'_r by the mean of Z''_i 's (in the first estimator) instead of just Z''_i (in the second estimator) lowers the variance because the mean has lower variance.

Exercise 4

We want to estimate $f(EW, EW^2) = (EW, EW^2 - (EW)^2)^T$ for some r.v. W . Let $Z := (W, W^2)$. Then $f(z) = (z_1, z_2 - z_1^2)^T$ where $z = (EZ(1), EZ(2)) = (EW, EW^2)$. Sample $W_1, \dots, W_R$ iid from W . Then we obtain estimator $f(\hat{z}_R) = (\frac{1}{R} \sum_{r=1}^R W_r, \frac{1}{R} \sum_{r=1}^R W_r^2 - (\frac{1}{R} \sum_{r=1}^R W_r)^2)^T$ of $f(z)$.

By weak LLN, $\bar{W}_R \xrightarrow{P} EW$ and $\bar{W}_R^2 \xrightarrow{P} EW^2$. Hence, $\hat{z}_R := (\bar{W}_R, \bar{W}_R^2)^T \xrightarrow{P} (EW, EW^2)^T =: z$. By continuous mapping theorem, $f(\hat{z}_R) \xrightarrow{P} f(z)$.

Let $Z_r = (W_r, W_r^2)^T$ iid in $\mathbb{R}^2$. By the multivariate CLT, where

$$\Sigma = Cov(Z) = \begin{bmatrix} EW^2 - (EW)^2 & EW^3 - EW EW^2 \\ EW^3 - EW EW^2 & EW^4 - (EW^2)^2 \end{bmatrix},$$

$$\sqrt{R}(\hat{z}_R - z) \rightsquigarrow \mathcal{N}_2(0, \Sigma).$$

The delta method yields that $\sqrt{R}(f(\hat{z}_R) - f(z)) \rightsquigarrow \mathcal{N}_2(0, f'_z \Sigma (f'_z)^T)$, where

$$\begin{aligned} f'_z \Sigma (f'_z)^T &= \begin{bmatrix} 1 & 0 \\ -2EW & 1 \end{bmatrix} \Sigma \begin{bmatrix} 1 & -2EW \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} EW^2 - (EW)^2 & 2(EW)^3 + EW^3 - 3EW EW^2 \\ -3EW EW^2 + 2(EW)^3 + EW^3 & -4(EW)^4 - 4EW EW^3 + 8(EW)^2 EW^2 + EW^4 - (EW^2)^2 \end{bmatrix}. \end{aligned}$$

Therefore, $f(\hat{z}_R) \rightsquigarrow \mathcal{N}_2(f(z), \frac{f'_z \Sigma (f'_z)^T}{R})$.

Exercise 5